

Module 3 Programming (Java Swing Project)

 Published

Challenge: Build a Simple Calculator

Description:

Your task is to create a basic calculator application that allows users to perform addition, subtraction, multiplication, and division operations. The calculator should have a graphical user interface (GUI) using Java Swing.

Skeleton code:

```
-----

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class CalculatorApp {
    // Define instance variables for GUI components (buttons, text fields, etc.)

    public CalculatorApp() {
        // Initialize the GUI components, set layouts, and add them to the frame.

        // Add action listeners to the buttons for performing calculations.

        // Display the frame.
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            new CalculatorApp();
        });
    }
}
```

```
}  
}
```

Grading Rubric (100 points):

1. GUI Layout (20 points):

- [10 points] The GUI includes buttons for digits (0-9).
- [5 points] The GUI includes buttons for addition (+), subtraction (-), multiplication (*), and division (/).
- [5 points] Proper layout and alignment of GUI components.

2. Functionality (40 points):

- [10 points] The calculator can perform addition correctly.
- [10 points] The calculator can perform subtraction correctly.
- [10 points] The calculator can perform multiplication correctly.
- [10 points] The calculator can perform division correctly.

3. Input Handling (20 points):

- [10 points] The calculator handles user input correctly and displays it in the input field.
- [10 points] Clear button or functionality to reset the calculator.

4. Error Handling (10 points):

- [10 points] The calculator gracefully handles division by zero and displays an appropriate message.

5. User Experience (10 points):

- [5 points] The GUI is user-friendly and easy to use.
- [5 points] Adequate feedback and messages for users (e.g., error messages, operation confirmation).

6. Code Quality (10 points):

- [5 points] Code is well-organized, follows Java naming conventions, and uses meaningful variable names.
- [5 points] Proper use of Java Swing library for creating the GUI.

Additional Information:

- The GUI can be created using Swing components like `JFrame`, `JPanel`, `JButton`, `JTextField`, etc.
- Ensure that the calculator performs calculations accurately, handles user input, and provides meaningful feedback to the user.

Points 100

Submitting a file upload

Due	For	Available from	Until
Sep 24	Everyone	-	Oct 22 at 11:59pm